Md. Abu Farhad

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Work Experience

Rooya
London, United Kingdom

Software Engineer
May 2025 - Current

- Built and scaled microservices for **AI-driven telematics systems** handling high-throughput real-time data pipelines.
- Implemented **DB read/write splitting** in PostgreSQL, improving query throughput by **~60%** and reducing read latency under peak load.
- Optimized core services with Go concurrency patterns, achieving **~40%** reduction in response latency and **~35%** gain in throughput.
- Moved driver identification into an async Kafka pipeline with AWS SageMaker, cutting trip-end processing time by **~60%**.
- Drove system reliability improvements, reducing downtime incidents by **~50%** across production services.

Relisource
Boston, United States

Software Engineer
Aug 2024 - June 2025

- Built a **Real-Time Visibility Platform (RTVP)** on large-scale logistics data, delivering actionable insights and reducing reporting latency by **~45%**.
- Designed ETL pipelines in **Databricks** to ingest, clean, and transform **10M+** records, cutting data processing time by **~50%**.
- Developed and deployed multiple **Azure Functions** to automate data workflows, eliminating **~30%** of manual data handling effort.

Monstarlab Bangladesh (MLBD)
Dhaka, Bangladesh

Software Engineer - II
March 2022 - July 2024

- Built high-performance microservices for **CAFU** (a world-leading fuel delivery and vehicle services platform), enhancing scalability and supporting over **~95k+** daily active users.
- Architected back-end services using **Golang** within an **event-driven architecture** to power the CAFU consumer app.
- Built the backend for **CAFU's** "Phantom" home screen in Go, serving personalized content to the app's main entry point; daily new users grew 30% after launch.
- Reduced error rate by **90%** compared to the legacy car-service system at **CAFU**.
- Integrated **gRPC and Protocol Buffers** to establish robust, lightweight communication across microservices.
- Built and maintained a **chat service** to enhance real-time user interaction and engagement.

Vivasoft Limited
Dhaka, Bangladesh

Software Engineer
April 2021 - February 2022

- Architected and built **15+** microservices from scratch using clean architecture principles, reducing onboarding time for new engineers by **~40%**.
- Containerized multiple project backends and database services using Docker, cutting deployment time by **~60%** and eliminating environment inconsistencies.
- Developed complex **REST API** features in a structured, testable manner, maintaining long-term code quality across a growing codebase.
- Automated the full integration and unit testing pipeline using Docker and shell scripts, achieving **~85%** test coverage and reducing manual QA effort significantly.

MyMedicalHUB
Dhaka, Bangladesh

Machine Learning Engineer
May 2020 - July 2020

- Built an **automated vertebrae detection and localization system** from CT scan images using a two-stage CNN architecture, achieving **89%** accuracy.
- Built end-to-end data pipelines for feature extraction and model training, reducing preprocessing time by **~65%** and enabling **3x faster** model iteration cycles.

Skills

Programming Languages	**Go** (Fluent), **Python** (Fluent), **C#** (Fluent), **C++** (Fluent), **Java** (Proficient) **Javascript** (Proficient)
Frameworks	Gin (Fluent), Echo (Fluent), ASP.NET Core (Proficient), FastAPI (Proficient).
Tools & Technologies	Docker, **Kubernetes** (Intermediate), Git, OOP, gRPC, GraphQL, RabbitMQ, Kafka, Terraform , CI/CD, Agile, etc.
Database	MySQL, PostgreSQL, CosmosDB, MongoDB, Redis, etc.
Cloud	**AWS** (Compute(EC2, Lambda), Networking(VPC, Route 53), Containerization(ECS, EKS), Messaging(SQS, SNS)), **Azure** (Service Bus, Stream Analytics, Function App, Event Hubs), etc.
Data Engineering	Databricks, Apache Spark, Azure (Data Lake, Data Factory), etc.
Research Skills & Tools	Machine Learning, Deep Learning, Keras, NumPy, Pandas, etc.

Achievements

Dec - 2019	**9th Position**, Inter-University Programming Contest	*VU, Bangladesh*
Aug - 2019	**3rd Runners-up**, Intra-University Programming Contest	*RUET, Bangladesh*
Apr - 2022	**1st Runners-up**, Monstarlab Global Hackathon	*Dhaka, Bangladesh*
Till now	**1700+**, Problems solved in different **online judges** →	

Projects

Real time visibility platform (RTVP)

United States

Client Based Project

Aug 2024 - May 2025

- A real-time visibility platform leveraging data engineering to provide actionable insights and optimize decision-making processes.
- Designed and implemented scalable data pipelines to enable real-time tracking, monitoring, and reporting across distributed systems.

Car Service

Dubai, UAE

Client Based Project

Oct 2022 - Jan 2024

- An **on-demand car service platform** covering a range of vehicle maintenance needs, including car wash, oil change, battery replacement, and tire change.
- **Tech Stack -** Microservice architecture, Golang, PostgreSQL, Goose, AWS S3, SNS.

WellTeam →

Adelaide, Australia

Client Based Project

Oct 2022 - Feb 2023

- A health and wellness platform delivering personalized nutrition plans and fitness coaching through structured online programs.
- Followed **Microservice** architecture with **Golang** as the primary language and **MySQL** as the database, ensuring a modular and maintainable codebase.

GoMail →

Dhaka, Bangladesh

Personal Project

Apr 2025 - March 2026

- A lightweight, terminal-based email client written in Go, drawing inspiration from tools like mutt, aerc, and himalaya.
- **Key Features -** TUI interface, IMAP & SMTP support, Clean and Modular Architecture, structured Logging, and robust Error Handling.

Golang Boilerplate →

Dhaka, Bangladesh

Personal Project

Aug 2022 - May 2024

- A production-ready **Clean Architecture** boilerplate in Golang, designed to serve as a solid starting point for building maintainable and scalable applications. It enforces clear separation of concerns and promotes best practices throughout the entire development lifecycle.
- **Key Features -** Clean Architecture Principles, Modularity, Dependency Injection, Database Integration, Environment Configuration, Structured Logging, and robust Error Handling.

Research Experience

Rajshahi University of Engineering & Technology (RUET)

Rajshahi, Bangladesh

Research Assistant

Dec 2019 - Feb 2021

- **Publication:** *Attention-Enhanced U-Net with Boundary-Aware Hybrid Loss for Low-Grade Glioma Segmentation*, IEEE BECITHCON 2026 (Accepted, Paper ID 307).
  - Proposed an attention-enhanced U-Net with a boundary-aware hybrid loss for precise low-grade glioma segmentation on MRI images.
  - Improved model performance, achieving a positive predictive value (PPV) of 0.87 and sensitivity of 0.89.

Education

Rajshahi University of Engineering & Technology (RUET)

Rajshahi, Bangladesh

B.Sc. in Computer Science and Engineering

Jan 2016 - Dec 2020

- **CGPA: 3.41/4.00**